



UnityCloud In-Service Software Upgrade Guide

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Affirmed Networks Inc.

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Release Abstract

This document contains a comprehensive guide to understand and use the ISSU feature for the various components.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. You can access Affirmed Networks Guides on the [Manuals](#). From here, you can also download a PDF version of an individual guide, or the full manual set for a product. For access and instructions, please contact Support.

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Introduction

In-Service Software Upgrade allows end users to upgrade or rollback the software of the components without affecting the service availability or performance. ISSU (In Service Software Upgrade) covers many components such as Observability, Networking, Database, Infra, and Applications SMF, UPF, and AMF. This content explains the impact, design, and strategy of ISSU for each component. Other content includes FAQ (frequently asked questions) and troubleshooting tips.

ISSU Orchestration

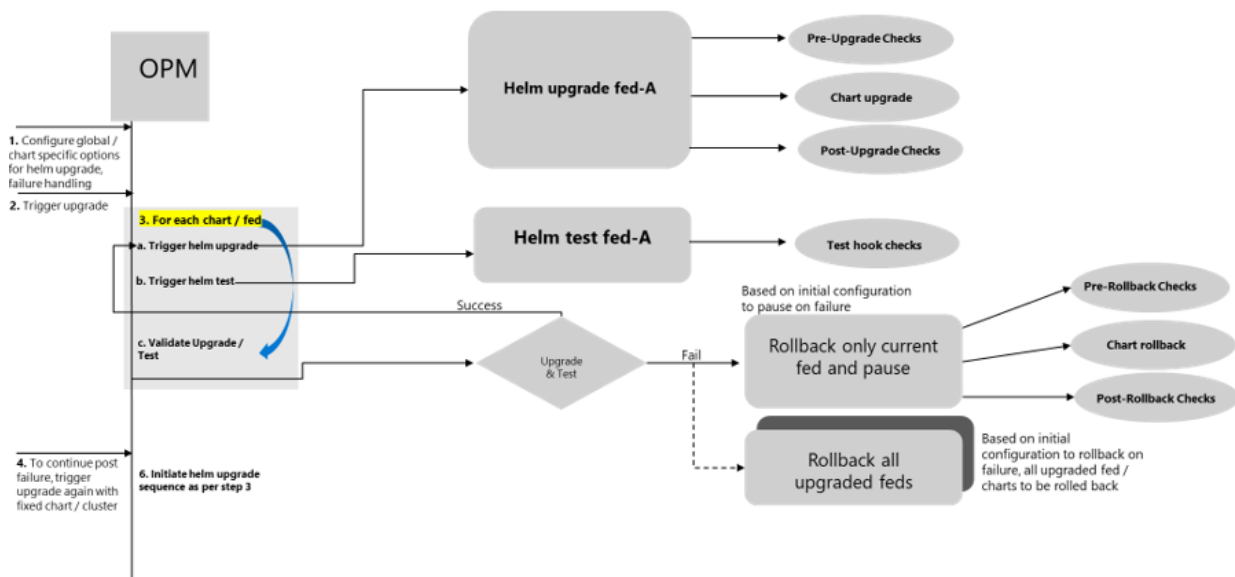
Upgrade is coordinated by the OPM or other packaging and deployment solutions which trigger the helm commands. Across the top-level network functions, the upgrade order is followed to trigger the upgrade. The upgrade order is dynamically generated based on federations present on the cluster and their known dependencies. The helm upgrade also triggers the pre and post upgrade and similarly for rollback, pre and post rollback hooks, which are executed to ensure pre and post check during upgrade / rollback are satisfied.

There are two failure policies that can be set:

- PauseOnFailure
- RollbackOnFailure

If the upgrade failure policy is set to pause-on-failure, upon hitting an upgrade failure, the orchestrator rolls back ONLY the particular federation which has failed to the previous stable release and pauses there. No further upgrades can be done. If the upgrade failure policy is set to rollback-on-failure, upon hitting a upgrade failure, all the upgraded federations are rolled back to their previous stable releases.

Figure 1. ISSU Orchestration



Note: The helm upgrade command should not be run separately from command line as the hooks will also be executed as part of it.

ISSU Impact and Design

ISSU has different impacts and design changes depending on the component. This content describes the main aspects of ISSU for each component, as well as the helm variables that are introduced or modified for ISSU.

Observability

ISSU for Observability allows users to upgrade or rollback the logging, monitoring, and tracing components without affecting the data collection or visualization. ISSU for Observability also uses helm hooks to perform pre and post checks for the upgrade and rollback, such as verifying the pod readiness, the data integrity, and the service availability.

Generic Assumptions for In-Service Upgrade

- The Kubernetes cluster is in a good state.
- Pods of the federation targeted for In-Service Upgrade are in a healthy state.

Recommendation & Limitation

fed-prometheus

- Recommendation:
 - fed-prometheus should be installed with persistent storage. So that we do not lose any historical data during the upgrade.
 - Local-path Provisioner is one of the recommended storage option which is easy to configure but it has the limitations mentioned below.
- Limitation:
 - If in-cluster (local storage) is used then during the upgrade all the historical data will be lost.
 - If localpathprovisioner is used as a storage option then data is persistent only if pod comes up on the same node. If the pod comes up on a different node, then data is not retained.

fed-elastic

- Recommendation:
 - fed-elastic must be installed with persistent storage either an external storage or localpathprovisioner must be configured.
- Limitation:
 - Currently, fed-elastic supports only upgrade. Fed-elastic can't be downgraded, which means rollback can't be supported to move back to the previous version. Let's say we perform version upgrade from 7.17.10-xpack to 7.17.13-xpack. Now for some reason we want to move back to 7.17.10-xpack, This is not possible to perform with Helm rollback command. We have to do a fresh installation of fed-elastic with 7.17.10-xpack.

Note: elastic replicates the data among the elasticsearch data nodes. Even with the localpathprovisioner configured there won't be any data loss.

fed-kafka-strimzi

- Recommendation:
 - fed-kafka-strimzi must be installed with persistent storage either an external storage.
 - The MirrorMaker component under the pod-kafka_strimzi is optional and disabled by default. It only needs to be enabled when you require data migration from an existing Kafka cluster.
- Limitation:
 - If localpathprovisioner is used as a storage option and storage type is selected as local-pv, then data is persistent only if pod comes up on the same node. If the pod comes up on a different node, then data is not retained.

Note: kafka replicates the data among the brokers. Even with the storage type local if we have more than 1 broker pods.

fed-kube-prom

- Recommendation:

- fed-kube-prom should be installed with persistent storage. So that we do not lose any historical data during the upgrade.
- Local-path Provisioner is one of the recommended storage option which is easy to configure but it has the limitations mentioned below.
- Limitation:
 - If in-cluster (local storage) is used then during the upgrade all the historical data will be lost.
 - If localpathprovisioner is used as a storage option then data is persistent only if pod comes up on the same node. If the pod comes up on a different node, then data is not retained.

Components

fed-elastalert

Mandatory fields to configure before upgrade

Table 1. Mandatory fields to configure before upgrade – fed-elastalert

Parameter
mesh.istio.egressGwTLS.enabled
mesh.istio.egressGwTLS.userProvided.cacert
mesh.istio.egressGwTLS.userProvided.tls-cert
mesh.istio.egressGwTLS.userProvided.tlskey
fileserver.nfs.remoteHost
pod-elastalert.paasSelectorElasticsearchName
pod-elastalert.receivers[0].server
pod-elastalert.receivers[0].port

Note: By default, this fed is configured for Halo-E integration support. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration. Additionally, please note that the same flag is present in fed-paas-helpers as well, so the configuration of these flags should be aligned, e.g. if Halo-E integration is turned off in fed-paas-helpers but NOT turned off while upgrading this fed, then there would be issues.

fed-jaeger

Mandatory fields to configure before upgrade

Table 2. Mandatory fields to configure before upgrade – fed-jaeger

Parameter
pod-jaeger.storage.elasticsearch.host

Note: By default, this fed is configured for Halo-E integration support. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration. Additionally, please note that the same flag is present in fed-paas-helpers as well, so the configuration of these flags should be aligned, e.g. if Halo-E integration is turned off in fed-paas-helpers but NOT turned off while upgrading this fed, then the GUI(s) in this fed would NOT be accessible.

fed-kafka-strimzi

The upgrade of kafka strimzi operator to a newer version, creates another controller/broker pods/LB service. While upgrading reuse the helm values for loadBalancerIP so the new LB services gets the same IP.

- Check if the loadBalancerIPs parameter is set to be same shared IP as the existing external bootstrap and Lb ip per broker service

Warnings

- If storage type is set to "local", the data of kafka controller will be lost.
- Strimzi kafka doesn't allow change of storage type in upgrade, so change in storage type will not reflect and never allowed.

fed-kube-prom

Mandatory fields to update before upgrade

Table 3. Mandatory fields to configure before upgrade – fed-kube-prom

Parameter
global.grafanaDatasource.url

Warnings

- If pvc not attached, the data of prometheus and grafana, will be lost.
- If localpathprovisioner enabled, after pod upgrade, pods comes on different node the data will be lost.

Note: By default, this fed is configured for Halo-E integration support. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration.

Additionally, please note that the same flag is present in fed-paas-helpers as well, so the configuration of these flags should be aligned, e.g. if Halo-E integration is turned off in fed-paas-helpers but NOT turned off while upgrading this fed, then the GUI(s) in this fed would NOT be accessible.

fed-kubedb-operator

This upgrade of fed-kubedb-operator to a newer version. Ensure the new elasticsearch version CRO (esversion) gets created after the upgrade. This is a mandatory for fed-elastic upgrade.

fed-elastic

KubeDB supports only version upgrades of deployed elasticsearch cluster without disrupting the service operations.

Upgrade

Note: For ISSU support, fed-elastic should be deployed with localpathprovisioner/external storage.

Note: Currently, fed-elastic supports only upgrade. Fed-elastic can't be downgraded, which means rollback can't be supported to move back to the previous version. Let's say we perform version upgrade from 7.17.10-xpack to 7.17.13-xpack. Now for some reason we want to move back to 7.17.10-xpack, This is not possible to perform with Helm rollback command. We have to do a fresh installation of fed-elastic with 7.17.10-xpack.

Note: By default, this fed is configured for Halo-E integration support. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration.

Additionally, please note that the same flag is present in fed-paas-helpers as well, so the configuration of these flags should be aligned, e.g. if Halo-E integration is turned off in fed-paas-helpers but NOT turned off while upgrading this fed, then there would be issues.

fed-prometheus

This upgrade of prometheus is to a newer version. While upgrading, reuse the helm values for loadBalancerIP so the new LB service gets the same IP.

Warnings

- If pvc not attached, the data of prometheus and grafana, will be lost.
- If localpathprovisioner enabled, after pod upgrade, pod comes up on different node the data will be lost.

Note: By default, this fed is configured for Halo-E integration support. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration.

Additionally, please note that the same flag is present in fed-paas-helpers as well, so the configuration of these flags should be aligned, e.g. if Halo-E integration is turned off in fed-paas-helpers but NOT turned off while upgrading this fed, then the GUI(s) in this fed would NOT be accessible.

fed-otel_operator

This upgrade of otel-operator to a newer version. While upgrading reuse the helm values for loadBalancerIP so the new LB service gets the same IP.

fed-otel_collector

This upgrade of otel-collector to a newer version.

Note: By default, this fed is configured for Halo-E integration support. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration.

Additionally, please note that the same flag is present in fed-paas-helpers as well, so the configuration of these flags should be aligned, e.g. if Halo-E integration is turned off in fed-paas-helpers but NOT turned off while upgrading this fed, then there would be issues.

fed-kibana

This upgrade of kibana is to a newer version. While upgrading, reuse the helm values for loadBalancerIP so the new LB service gets the same IP.

Table 4. Mandatory fields to configure before upgrade – fed-kibana

Parameter
pod-kibana.elasticExtIP

Note: By default, this fed is configured for Halo-E integration support. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration. Additionally, please note that the same flag is present in `fed-paas-helpers` as well, so the configuration of these flags should be aligned, e.g. if Halo-E integration is turned off in `fed-paas-helpers` but NOT turned off while upgrading this fed, then the GUI(s) in this fed would NOT be accessible.

fed-grafana

This upgrade of grafana is to a newer version. While upgrading, reuse the helm values for loadBalancerIP so the new LB service gets the same IP.

Note: By default, this fed is configured for Halo-E integration support. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration. Additionally, please note that the same flag is present in `fed-paas-helpers` as well, so the configuration of these flags should be aligned, e.g. if Halo-E integration is turned off in `fed-paas-helpers` but NOT turned off while upgrading this fed, then the GUI(s) in this fed would NOT be accessible.

fed-paas-helpers

This upgrade of fed-paas-helpers is to a newer version. While upgrading, reuse the helm values for loadBalancerIP so the new LB service gets the same IP.

Note: While upgrading `fed-paas-helpers`, please refer to the section “Halo-E Integration with UC GUIs” in the Deployment Guide to properly integrate with Halo-E. In case of any missing configuration, the `fed-paas-helpers` upgrade will fail. If Halo-E integration is NOT required to be enabled, then please configure `global.authType = "local"` to turn off Halo-E integration.

Helm Hooks

fed-elastalert:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state, check for elastalert health using fqdn.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-jaeger:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-kafka-strimzi:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-kube-prom:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state check for prometheus metrics through fqdn api and grafana health api.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state check for grafana api health.

fed-kubedb-operator:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-elastic:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-prometheus:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state, check for prometheus metrics through fqdn api.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-otel-operator:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-otel-collector:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state, check for otel conf through logs.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-kibana:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state.

fed-grafana:

- Pre-upgrade: Check if all pods are in ready state and n/n state.
- Post-upgrade: Check if all pods are in ready state and n/n state, check for grafana health api.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state, check for grafana health api..

fed-paas-helpers:

- Pre-upgrade: Check if all pods are in ready state.
- Post-upgrade: Check if all pods are in ready state and n/n state, check for fluentd conf and grafana api health.
- Pre-rollback: NA
- Post-rollback: Check if all pods are in ready state and n/n state, check grafana health api.

Networking

ISSU for Networking allows users to upgrade or rollback the networking components of Istio which include Istio control plane and data plane comprising of ingress and egress gateways, without affecting the network connectivity or security. ISSU for Networking uses a rolling update strategy, which means that the pods are updated one by one with the new version, while the old pods are still serving the traffic. ISSU for Networking also uses helm hooks to perform pre and post checks for the upgrade and rollback, such as verifying the pod readiness, the network configuration, and the service availability.

There are three basic resources in fed-istio:

- Operator deployment
- Istiod replicaset
- Istio-cni daemonset

Out of this, only the operator is actually deployed by helm, it in-turn controls istiod and istio-cni, as well as the istio gateway resources in other feds.

Operator also controls various istio related crds.

fed-istio has postDelete, preinstall, postInstall, preUpgrade, postUpgrade and postDelete helm hooks, for ISSU only the Upgrade and Rollback hooks are relevant.

- preUpgrade – checks current fed-istio is in good state before triggering upgrade, ie checks all pods are N/N in fed-istio namespace.
- postUpgrade – blocks till fed-istio is in good state after helm upgrade, ie till istio operator, istiod, istio-cni pods are created and go to N/N.

- postRollback – blocks till fed-istio is in good state after helm rollback. Same check as postUpgrade.

Database

Resource Description

Redis deployment is driven by the following three components:

1. fed-redis-cluster - The fed-redis-cluster pods constitute the Redis cluster and hold all data. The time taken to upgrade is high because existing data needs to be migrated.
2. fed-redis-operator - The fed-redis-operator manages the Redis cluster. Only one of the three fed-redis-operator pods is the leader and actively manages the cluster. A rolling upgrade of the fed-redis-operator ensures that there is always a leader during the upgrade, which watches over and manages the cluster.
3. udsf-mgmt-op service (running as part of fed-paas-helpers) - The udsf-mgmt-op service watches over and publishes database status to be consumed by other services. It is also responsible for installing search indices. However, since it continuously monitors the database and changes in the indices, it can still handle any events that occur while it is being replaced during an upgrade.

Helm Hooks

fed-redis-operator:

- Pre-upgrade: Check if there is at least one operator pod ready.
- Post-upgrade: Check if all pods for the operator deployment are ready.
- Pre-rollback: Check if there is at least one operator pod ready.
- Post-rollback: Check if all pods for the operator deployment are ready.

fed-redis-cluster:

- Pre-upgrade: Check if the "RedisCluster" state is "OK".
- Post-upgrade: Check if the "RedisCluster" state is "OK" and if the expected number of primaries and replicas are present.
- Pre-rollback: Check if the "RedisCluster" state is "OK".
- Post-rollback: Check if the "RedisCluster" state is "OK" and if the expected number of primaries and replicas are present.
- HelmTest (Runs post-upgrade): Check if data could be stored/read from all shards.

Infra

ISSU for Infra allows you to upgrade or rollback the infra components such as Kubernetes, Docker, Helm, and Istio without affecting the cluster stability or performance. ISSU for Infra uses a node-by-node strategy, which means that the nodes are updated one by one with the new version, while the other nodes are still serving the traffic. ISSU for Infra also uses helm hooks to perform pre and post checks for the upgrade and rollback, such as verifying the node readiness, the cluster health, and the service availability.

Configuration Manager

Changes in configuration manager:

- **Upgrade support:** cfgmgr supports upgrade of schema.
- **Versioned Keys:** Use of 'versioned' etcd key for upgrade and rollback support and distribution of configuration based on schema at use.
- **Failure Scenarios:** Handling of failure scenarios during upgrade and rollback with a design Change to track the LCM states (upgrade or rollback) to prevent schema downgrade or upgrade during pod restart or non-upgraded or non-downgraded pods.
- **Application Rules:** Application teams must follow guidelines for schema changes for seamless upgrade and downgrade.

SMF

ISSU for SMF allows you to upgrade or rollback the SMF component without affecting the session management or the user plane function. ISSU for SMF uses a stateful set strategy, which means that the SMF pods are updated one by one with the new version, while the old pods are still serving the traffic. ISSU for SMF also uses helm hooks to perform pre and post checks for the upgrade and rollback, such as verifying the pod readiness, the session state, and the service availability.

SMF supports pre and post upgrade/rollback hooks to validate and track the status of ISSU.

Helm Hooks in SMF

- Pre-Upgrade hook:
 - Checks pod-status, NF monitor status and KPI metrics to report the state of SMF federation for upgrade.
 - These metrics can also be configurable to enable the hook to make informed decisions about proceeding with the upgrade.
- Post-upgrade: Checks the upgrade status through the validation of metrics to determine if upgrade can continue based on:

- Pod Status
- NF Status
- KPI monitoring
- Pre-rollback: Reports state of SMF Federation prior to rollback.
- Post-rollback: Reports status or rollback for any failures.

Graceful shutdown is not supported on most microservices in SMF federation. Only smfcc microservice supports graceful shutdown. In flight transactions for call processing may fail depending on which microservices are being upgraded during the transaction during ISSU. This may result in session setup failures and/or session loss with session modification failures during ISSU.

Hooks Support

The following configuration defined in values.yaml for ISSU support:

- Pre Upgrade Hook:
 - validate: True (when enabled failure will result in action rollback/pause)
 - reporting: # default to generate report during pre-upgrade validation (this overrides outcome of validate action)
 - nfMonitor: True # default is to generate report of NF monitor status.
 - pod: True # default is to generate report of pod status.
 - Kpi: True # default to generate report of KPI metrics.
- Post Upgrade Hook:
 - validate: True (when enabled failure will result in action rollback/pause)
 - nfMonitor:
 - validate: true
 - pod:
 - name: xxx (ex: smfcc)
 - validate: true
 - value: 100 (% of pods that should be up for success)
 - Monitoring KPIs

Monitoring KPIs

smfcc_nsmf_pdu_session_update_sm_context_acceptable_failures
smfcc_gtpc_modify_bearer_request_acceptable_failures
smfcc_an_release_acceptable_failures
smfcc_service_request_acceptable_failures
smfcc_xn_handover_acceptable_failures
smfcc_ue_deregistration_acceptable_failures
smfcc_pgw_ue_initiated_detach_acceptable_failures
smfcc_ue_initiated_pdu_session_release_acceptable_failures
smfcc_pgw_modify_bearer_request_acceptable_failures
smfcc_pcf_initiated_policy_modification_acceptable_failures
smfcc_pgw_pcf_initiated_policy_modification_acceptable_failures

- Threshold and enable options per KPI

Parameter	Value	Description
kpi_min_metric_value	100	Fed SMF level - KPI will be calculated only if metric has minimum configured value.
kpi_sampling_interval	60	Fed SMF level - Duration for KPI monitoring in second during upgrade post PODS are in n/n state.
Validate	True	per KPI - If false, then upgrade will not be failed even KPI does not match passing criteria.
Value	60	per KPI threshold - KPI to be compared against maximum value %.

SMF supports 3GPP standard version reporting to NRF for SMF services such as *nsmf-pdusession* and *nsmf-event-exposure with NF profile registration/updates*.

Yang configuration

- smf service service-profile nsmf-pdusession version 17.4.0
- smf service service-profile nsmf-event-exposure version 16.4.0

UPF**ISSU design changes**

- UPF does the rolling upgrade of PPE pods where 1 PPE pod will be taken out of service as the reset of the PPEs serving user traffic.
- It mainly uses PPE Active/Standby solution as part of the upgrade.

- Once K8 initiates upgrade of PPE, L3RM switches all the active resource blocks of that PPE to other PPEs and then only restarts the PPE.
- But if during PPE upgrade - if L3RM is already undergoing upgrade, PPE will continue to host the data traffic and control traffic (modify and delete Requests) until new L3RM comes up and switches all the active RBs of that PPE.
- After a new PPE is created with new image, it would come up as Standby PPE and all the active user records would be synced to this new PPE.
- After syncing all the user session records, L3RM would switch the resource blocks to Active.
- Only after completing syncing of data records, this PPE would go into the ready state and this allows K8 to initiate upgrade on the another PPE.

Helm variable introduced and their description for ISSU

N/A

Hooks

- Post Upgrade script for fed-upf is introduced which ensures all the pods are in n/n state and then only replies back that upgrade is success. (PreUpgrade script already exists as part of 4.3)
- Prestop and poststart hooks are introduced in L3RM and PPE pod, which helps in sequencing these pods.

AMF

ISSU for AMF allows users to upgrade or rollback the AMF component without affecting the access and mobility management or the control plane function. ISSU for AMF uses a deployment strategy, which means that the AMF pods are updated with the new version, while the old pods are terminated. ISSU for AMF also uses helm hooks to perform pre and post checks for the upgrade and rollback, such as verifying the pod readiness, the control plane state, and the service availability.

MME

ISSU for MME allows seamless upgrading or rolling back of the Mobility Management Entity (MME) component without disrupting access, mobility management, or control plane operations. During an upgrade, new MME pods are deployed with the updated version, while old pods are gracefully terminated. Helm hooks are used for pre and post checks, including verifying pod readiness, assessing control plane status, and ensuring service availability. ISSU for MME ensures reliable maintenance while keeping the system up to date.

Helm Hooks in MME

- Pre-Upgrade hook:
 - Checks pod status, NF status to report the state of MME federation before upgrade.
- Post-Upgrade hook:
 - Checks pod status, NF status to report the state of MME federation after upgrade.
- Test hook:
 - Pod Status
 - NF Status
 - KPI monitoring
- Pre-rollback:
 - Checks pod status, NF status to report the state of MME federation before rollback. The output of pod status and NF status is just logged, the rollback is always allowed to succeed irrespective of this.
- Post-rollback:
 - Checks pod status, NF status to report the state of MME federation after rollback. The output of pod status and NF status is just logged, the rollback is always allowed to succeed irrespective of this.

Graceful shutdown is not supported in MME. In flight transactions for call processing may fail depending on which microservices are being upgraded during the transaction during ISSU and this may result in 3GPP procedure failures during ISSU.

Hooks Support

The following configuration is defined in values.yaml for ISSU support:

- timeout:
 - An integer value indicating the timeout in minutes for the upgrade hooks. This must be aligned with the overall ISSU timeout value.

Each of the hooks mentioned in the previous section are separated as preUpgrade, postUpgrade, test, preRollback and postRollback. They are provided with separate controls for each of the checks.

- validatePods:
 - This control is used to specify if the hook needs to validate pods. Takes values true/false.
- validateKpis:

- This control is used to specify if the hook needs to validate KPIs. Takes values true/false.
- validateNfState:
 - This control is used to specify if the hook needs to validate NF status. Takes values true/false.

Monitoring KPIs
mmecc_emm_attach_success_rate
mmecc_emm_attach_attempt_rate
mmes1_s1_incoming_message_rate

The thresholds for each of these KPIs are configurable in the deployment scripts.

ISSU Upgrade/Rollback Strategy

The ISSU upgrade strategy is the process of upgrading the software of the components from the old version to the new version using the ISSU feature.

Observability

- For the components where as service impacting and multiple replicas, rolling upgrade strategy is set-up.
- For the components which are singleton like operators or Deamonset for them recreate strategy is set-up.

Networking

Update / Rollback strategy is as follows for Istiod and istio-operator

Istiod

strategy: rollingUpdate:

maxSurge: 100%

maxUnavailable: 25%

Istio-operator

strategy: rollingUpdate:

maxSurge: 25%

maxUnavailable: 25%

Database

fed-redis-cluster

Pod deployment is managed by fed-redis-operator. Exact mechanism to do that is discussed in [ISSU Traffic Impact](#).

fed-redis-operator

strategy: Recreate

UPF

- No new additional PPE pods would be created during the upgrade and only one PPE pod will be taken out of service at any time during the upgrade.

ISSU postUpgrade Manual Validation

Please note that the following post-validation manual steps are optional and provided for users who wish to further validate their results after ISSU. Feel free to skip these steps if you are satisfied with the ISSU Upgrade.

Observability

fed-elastalert:

- All the pods in fed-elastalert namespace should be in a running state.
- SSH into fileserver.nfs.remoteHost and check if alert rules are present in /filestore/fed-elastalert/alerts
- If there are any existing alerts that were getting triggered before the upgrade, it should continue to trigger alerts for those logs.

fed-jaeger:

- Login to Jaeger query GUI page and check if you can access it..
- Verify if you can see the traces and spans in Jaeger GUI.
- All the pods in fed-jaeger namespace should be in a running state.

fed-kafka-strimzi:

- The Kafka cluster has to be in the "ClusterRunning" state. FYI, you will see it goes to states like "ClusterRollingUpgrading" and "ClusterReconciling" before becoming "ClusterRunning" again.
- Make sure all the pods in the namespace are in a running state. CruiseControl generally takes a little bit more time. Also when mirror maker is enabled, expect container restarts until Kafka brokers are up and running and it gets attached to it.

fed-kube-prom:

- Verify kube-prom's prometheus and alertmanager GUI is accessible.
- Verify if you can scrape from metrics by querying for any metrics in kube-prom's prometheus GUI.
- Verify kube-prom's grafana GUI is accessible and dashboards has data to be visualized.
- All the pods in the fed-kube-prom namespace should be in a running state.

fed-kubedb-operator:

- All the pods in the fed-kubedb-operator namespace should be in a running state.

fed-elastic:

- Use the following command to verify the status of elastic search.
[maint@platform-test-k8-master-1-a00061-4b2c09f7b029c1b8 ~]# kubectl get elasticsearch -n fed-elastic NAME VERSION STATUS AGE fed-elastic 7.9.3-searchguard Ready 10h

fed-prometheus:

- Check that all the pods in fed-prometheus are running.
- Open Prometheus UI search for any metric.
- Verify if alertmanager UI is accessible.

fed-otel-operator:

- Check all the pods in fed-otel_operator are running.

fed-otel-collector:

- Check all the pods in fed-otel_collector are running.
- Open Jaeger UI see all the spans are visible.

fed-kibana:

- Check that all the pods in fed-kibana are running.
- Open the Kibana UI see all the logs that are visible and check the indices in fed-elastic.
- Check if fluentd* and fluentd-event* default indices are present in kibana saved objects.

fed-grafana:

- Check that all the pods in fed-grafana are running.
- Open the grafana UI and view the dashboards and ensure graphs display without disruption.

fed-paas-helpers:

- Check that all the pods in fed-paas-helpers are running

ISSU Traffic Impact

The ISSU impact on traffic is the effect of the ISSU feature on the service traffic during the upgrade or rollback operation. The ISSU impact on traffic depends on the component and the strategy.

Observability

- No significant impact on traffic while performing ISSU's.

Networking

fed-istio resources themselves do not handle traffic, the traffic handling components are istio gateways and istio proxies in NF namespaces.

Istio gateways are reconciled by operator, they will restart if istio versions differ. It's expected that clients will reconnect to the new gateways automatically as service remains same, and since gateways themselves are stateless there is no traffic impact.

Changes in consistent hashing are expected to be handled by NF.

Proxies do not restart, they will be upgraded with NF upgrade, and they are stateless themselves, hence no impact.

Database

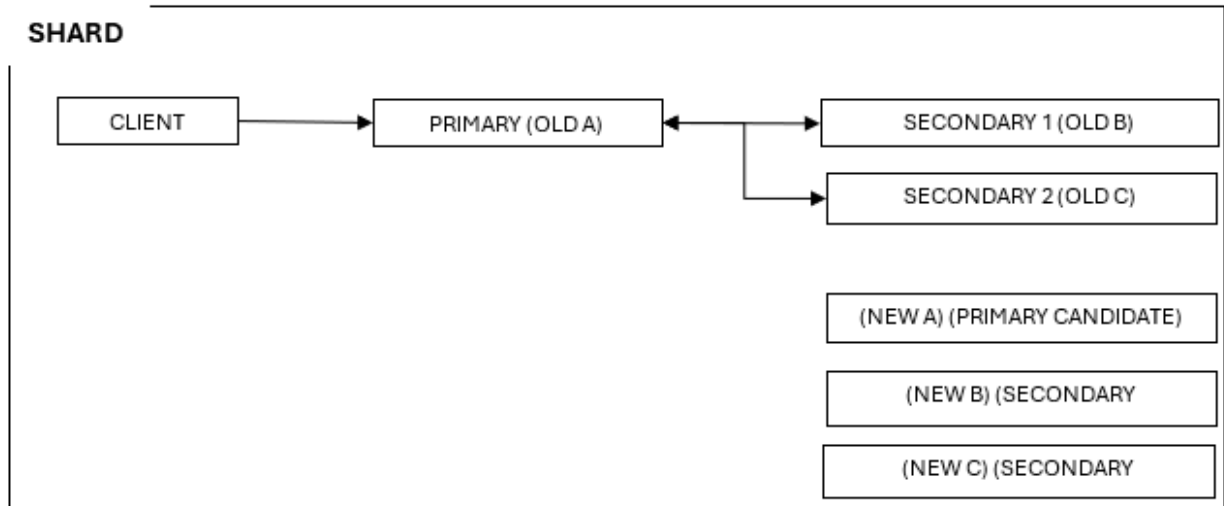
The pods running under fed-redis-cluster handle traffic. These pods are reconciled by the Redis operator running under fed-redis-operator.

The fed-redis-cluster pods are replaced only if the 'redisclusters' CRO under fed-redis-cluster is updated, i.e., during the upgrade of fed-redis-cluster.

To avoid any impact on traffic, the Redis operator replaces one shard at a time. The following strategy is followed:

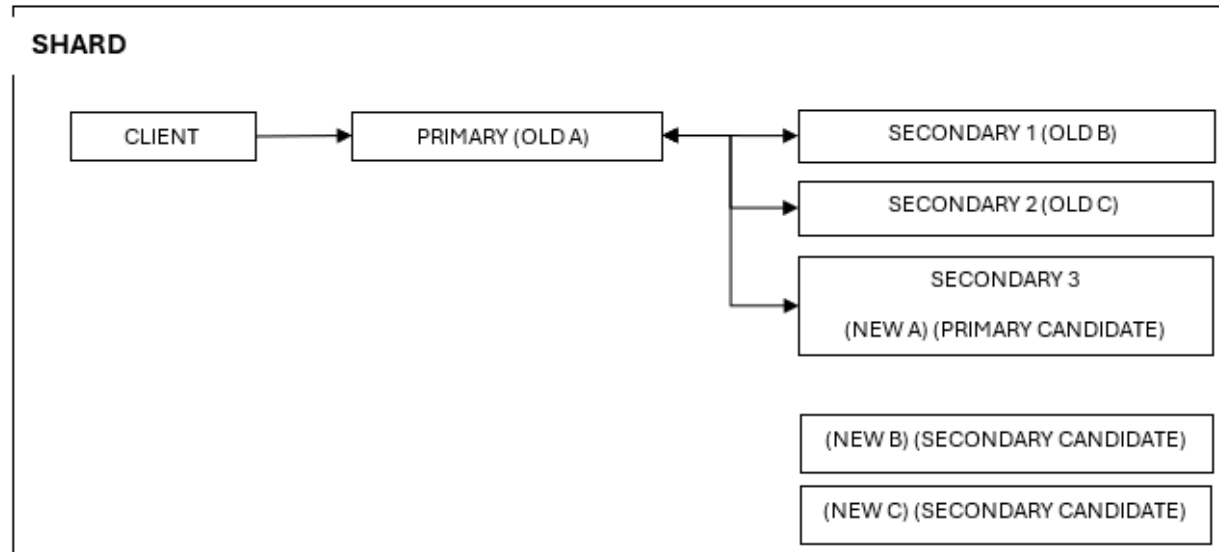
1. For the shard being replaced, create replicas of all primary and secondary Redis pods. At this moment, the shard primary pod is handling traffic.

Figure 2. SHARD – Step 1



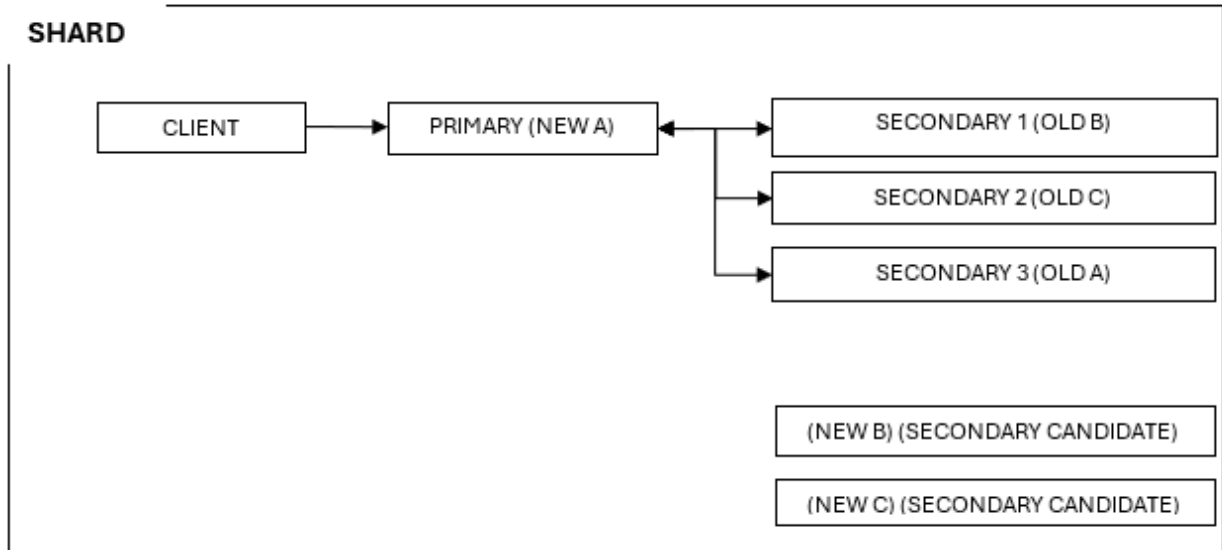
2. Add the new primary candidate as a standby pod to the existing primary. Wait for "New A" to get in sync with "Old A".

Figure 3. SHARD – Step 2



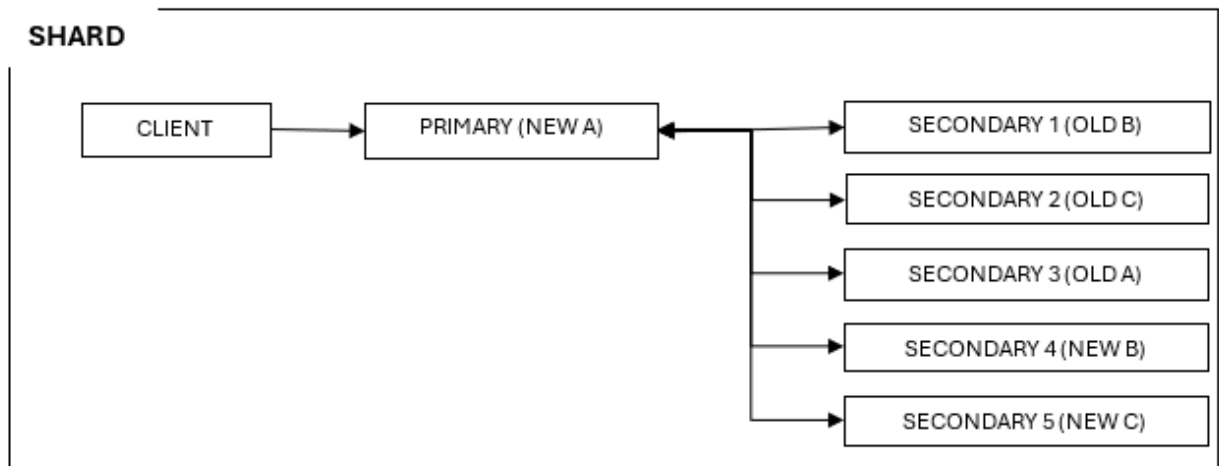
3. Perform a failover to make "New A" the primary.

Figure 4. SHARD – Step 3

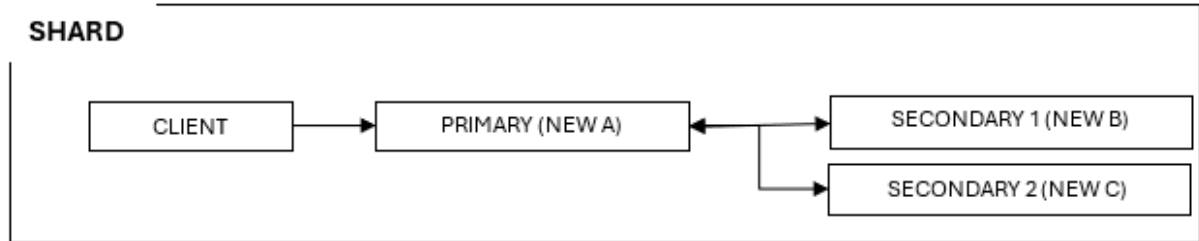


4. Add the remaining new nodes as secondaries and let syncing start.

Figure 5. SHARD – Step 4



5. Add the remaining new nodes as secondaries and let syncing start.

Figure 6. SHARD – Step 5

As indicated from the preceding sequence of steps, a Redis shard always remains in service during the upgrade and does not impact traffic.

UPF

Traffic impact during ISSU is:

- During upgrades when multiple services are being upgraded (for ex. L3RM or PFCP-U with PPE), the data loss and in-flight transaction failures observed may be more compared to when only PPE is being upgraded. The system will recover with peer NF/UR retries without any operator intervention.
- When the PPE is taken down (and when the services are reverted to a new pod instance), the data connectivity is preemptively switched to the secondary PPEs, however there will be windows where the in-flight transactions will timeout or fail resulting in session deletion - however these will be restored successfully when the peer NF or UE retries.

Infra

TBD

OSS Aspects

The following topics contain OSS (open-source software) information.

Networking

Istio project has a fast release cadence, roughly a major release every 5-6 months - <https://istio.io/latest/news/releases/>

Istio upgrade in unity cloud is driven by UC features and Microsoft OSS guidance.

Any UC release with istio upgrade release noted will be compatible with the previous N-1 fed-istio, unless explicitly noted as otherwise.

Database

A major Redis version is released every year, with a minor version released every six months.

Redis upgrades in Unity Cloud are driven by Unity Cloud features and Microsoft OSS guidance.

Any Unity Cloud release with a noted Redis upgrade will be compatible with the previous N-1 fed-redis-cluster, unless explicitly noted otherwise.

Observability

Table 5. Namespace Image information

Namespace	Image
Fed-elastic	• elasticsearch:7.17.13-patched -tag kubedb_elasticsearch • prometheuscommunity/elasticsearch-exporter:v1.6.0 -tag elasticsearch_exporter • tianon/toybox:0.8.10 -tag elasticsearch_toybox • kubedb/elasticsearch-init:7.17.10-xpack-v2023.07.14-patched -tag elasticsearch_init • untergeek/curator:7.0.0 -tag curator_7_0_0
Fed-elasticsearch	• bitsensor/elastalert:elastalert_3_0_0_beta_1
Fed-kibana	• kibana/kibana:7.17.13
Fed-jaeger	• mcr.microsoft.com/oss/jaegertracing/jaeger-query:v1.49.0 • mcr.microsoft.com/oss/jaegertracing/jaeger-collector:v1.49.0
Fed-prometheus	• mcr.microsoft.com/oss/prometheus/prometheus-operator:v0.69.1 • mcr.microsoft.com/oss/prometheus/prometheus:v2.48.0

Namespace	Image
	mcr.microsoft.com/oss/prometheus/prometheus-config-reloader:v0.70.0
Fed-kube-prom	- mcr.microsoft.com/oss/prometheus/prometheus:v2.48.0 - k8s-staging-prometheus-adapter/prometheus-adapter-amd64:v0.10.0 - mcr.microsoft.com/oss/prometheus/prometheus-operator:v0.70.0 - mcr.microsoft.com/oss/prometheus/node-exporter:v1.7.0 - mcr.microsoft.com/oss/kubernetes/kube-state-metrics:v2.10.1 - grafana/grafana:10.2.2 - mcr.microsoft.com/oss/prometheus/alertmanager:v0.26.0 - mcr.microsoft.com/oss/prometheus/prometheus-config-reloader:v0.70.0 - mcr.microsoft.com/oss/kubernetes/configmap-reload:v0.12.0 - mcr.microsoft.com/oss/kubernetes/kube-rbac-proxy:v0.15.0
Fed-otel_operator	- mcr.microsoft.com/oss/otel/opentelemetry-operator:v0.81.0 - mcr.microsoft.com/oss/kubernetes/kube-rbac-proxy:v0.15.0
Fed-otel-collector	mcr.microsoft.com/oss/otel/opentelemetry-collector-contrib:0.79.0
Cna-keycloak	keycloak/keycloak:22.0.1
Fed-grafana	grafana/grafana:10.2.2

FAQs and error scenarios

This section provides some frequently asked questions and general error scenarios related to the ISSU feature, as well as some possible solutions or recommendations.

Question: How can I check the status of the ISSU operation?

Answer: You can check the status of the ISSU operation by using the helm status command for the component. You can also check the pod status, the helm hook status, and the logs for the component.

Question: How can I cancel the ISSU operation?

Answer: You can cancel the ISSU operation by using the helm rollback command for the component. You can also disable the ISSU feature by setting the helm variables to false.

Question: What are the prerequisites for using the ISSU feature?

Answer: The prerequisites for using the ISSU feature are:

- The cluster must be healthy and stable.
- The backup must be taken and available.
- The helm charts must be ready and compatible.

Question: What are the risks or limitations of using the ISSU feature?

Answer: The risks or limitations of using the ISSU feature are:

- The ISSU feature may not support all the components or versions.
- The ISSU feature may not guarantee zero downtime or zero data loss.
- The ISSU feature may consume more resources or time.
- The ISSU feature may encounter errors or failures due to various reasons.

Question: What are some common errors or failures that may occur during the ISSU operation?

Answer: Some common errors or failures that may occur during the ISSU operation are:

- The helm upgrade or rollback command may fail due to syntax errors, network errors, or compatibility issues.
- The helm hooks may fail due to timeout errors, logic errors, or dependency issues.
- The pod status may be stuck in pending, creating, or terminating due to resource errors, configuration errors, or scheduling errors.

- The service availability may be degraded or disrupted due to network errors, security errors, or performance errors.
- The functionality may be impaired or corrupted due to data errors, state errors, or logic errors.

Question: What are some possible solutions or recommendations for resolving the errors or failures that may occur during the ISSU operation?

Answer: Some possible solutions or recommendations for resolving the errors or failures that may occur during the ISSU operation are:

- Check the helm status, the pod status, the helm hook status, and the logs for the component.
- Retry the helm upgrade or rollback command with the correct syntax, network, and compatibility.
- Increase the helm hook timeout or fix the helm hook logic or dependency.
- Allocate more resources or adjust the configuration or scheduling for the pod.
- Restore the backup or fix the network, security, or performance for the service.
- Verify the data, state, or logic for the functionality.

Question: What are some best practices or tips for using the ISSU feature?

Answer: Some best practices or tips for using the ISSU feature are:

- Plan the ISSU operation carefully and test it in a staging environment before applying it in a production environment.
- Follow the ISSU upgrade or rollback strategy and the steps as described in this document.
- Use the ISSU feature only when necessary and with caution.
- Monitor the ISSU operation closely and be ready to intervene or abort if needed.
- Report any issues or feedback to the ISSU team or the support team.

Upgrade Failures – what to do

In case of upgrade failure, the following steps are recommended:

1. Identify the cause and the scope of the failure by checking the helm status, the pod status, the helm hook status, and the logs for the component
2. Rollback the upgrade by using the helm rollback command for the component
3. Restore the backup if needed
4. Disable the ISSU feature by setting the helm variables to false
5. Contact the ISSU team or the support team for further assistance

Networking

Failed upgrade/rollbacks will leave a job with the same name in Error state in fed-istio namespace, it's logs will indicate the problem which failed upgrade.

- Problem – operator is taking long time to create resources.

Known Issue. Tail operator logs, it will show logs like “Processing ingress gateways”, sometimes operator deploys resources in wrong order and must timeout several times before proceeding. Depending on the feds deployed, this may take 5-10 minutes.

- Problem – pods with istio proxy are crash looping on init
 - This should not happen, but issues with istio-cni can result in pod networking being in a bad state... tail istio-cni logs and you should see logs “Pod detected as broken”.
 - Not other solution other than rollback of fed-istio
- Problem – NF upgrade/install fail with errors on istio related resources like VirtualServices, DestinationRule.
 - Again, should never happen, but this indicates that istio crds are in bad state after upgrade. Rollback fed-istio to fix.
- Problem – istio proxies/gateways are not learning ips of new pods
 - Should never happen. Indicates incompatibility in istio control plane, rollback fed-istio to fix.

Database

It is expected that the upgrade of fed-redis-cluster takes a long time (30-50 minutes).

Failed upgrades/rollbacks will leave a job with the same name in an Error state in the fed-redis-cluster/fed-redis-operator namespace; its logs will indicate the problem that caused the failed upgrade.

- Problem: fed-redis-cluster upgrade is not proceeding. New pods are stuck in a pending state.
 - Check for resources on the cluster. It could happen if the cluster is low on memory or CPU. Freeing up resources will allow the upgrade to proceed.
- Problem: fed-redis-cluster upgrade is not proceeding. None of the pods are in a pending state.
 - This should not happen, but if a new pod is not promoted to primary after a long time, check its logs. If any failures are observed, deleting (restarting) the pod will allow it to recover and start the sync again.
- Problem: UDSF client dashboard indicates that search requests are failing.
 - This should never happen, but it indicates that meta-schemas are lost on certain Redis pods. Identify the problematic primary Redis pods from the Redis server Grafana dashboard. Delete them one at a time to trigger a failover in the respective shards. This allows pods with correct meta-schemas to take over.